

Business Requirements Document

Retail Analytics — Databricks Lakehouse Use Case

Version: 1.1 — Updated with Medallion Architecture **Date:** 7 September 2026

1. Executive Summary

This document defines the business requirements for a Retail Analytics platform that consolidates customer, product, online order, store sales, inventory, and payment data into a governed Databricks Lakehouse. The solution follows the Medallion Architecture pattern — Bronze, Silver, and Gold — to progressively improve data quality and business usability.

The target outcome is a trusted, scalable analytics foundation supporting executive dashboards, Customer 360, inventory intelligence, machine learning, and AI-assisted business analysis.

2. Business Problem

Retail data is distributed across e-commerce, point-of-sale, CRM, inventory, payment, and file-based systems. Manual reconciliation and reporting create delays and inconsistent metrics. The organization needs a governed platform that integrates batch and near-real-time data while preserving source history and producing trusted business datasets.

3. Business Objectives

- Create a single source of truth for retail analytics.
- Automate ingestion and processing of customer, product, order, store-sales, inventory and payment data.
- Improve data quality, consistency, traceability and governance.
- Provide standardized executive and operational KPIs.
- Enable Customer 360 and RFM customer segmentation.
- Identify inventory shortages and replenishment risks proactively.
- Compare online and physical-store performance.
- Enable ML use cases such as churn prediction and sales forecasting.
- Enable AI-assisted analysis of business performance.

4. Medallion Architecture

The solution shall implement the Databricks Medallion Architecture. Data moves through progressively refined layers: **Bronze** → **Silver** → **Gold**. Each layer has a clear purpose and data contract.

End-to-end flow: Source Systems → Ingestion → Bronze → Data Quality & Transformation → Silver → Business Aggregation → Gold → BI / ML / AI / Applications

Layer	Purpose	Key Activities	Example Tables
Bronze	Raw/source-aligned data	Ingest, preserve source structure, add ingestion metadata, retain history, support replay/audit	bronze_customer, bronze_product, bronze_orders, bronze_store_sales, bronze_inventory, bronze_payment

Layer	Purpose	Key Activities	Example Tables
Silver	Trusted and integrated data	Validate, cleanse, deduplicate, standardize, enrich, resolve relationships, quarantine bad records	silver_customer, silver_product, silver_orders, silver_store_sales, silver_inventory, silver_payment
Gold	Business-ready data	Apply business rules, joins, aggregations, KPI logic and analytical models	gold_customer_360, gold_sales, gold_store_performance, gold_inventory_risk, gold_payment_analytics, gold_customer_segments

4.1 Bronze Layer Requirements

- Store source records with minimal transformation.
- Preserve raw values and ingestion timestamps.
- Support incremental ingestion and reprocessing.
- Maintain source-system identifiers for lineage.
- Do not silently discard invalid source records.

4.2 Silver Layer Requirements

- Apply schema and data-type validation.
- Remove or flag duplicate business transactions.
- Standardize dates, amounts, status codes and reference values.
- Validate customer/product/store relationships.
- Separate invalid records into a quarantine/error dataset.
- Produce trusted, reusable datasets for downstream consumers.

4.3 Gold Layer Requirements

- Implement approved business definitions and KPI calculations.
- Create dimensional/analytical datasets optimized for reporting.
- Provide Customer 360, sales, store, inventory and payment subject areas.
- Expose curated datasets to BI, ML and AI workloads.
- Maintain clear ownership and lineage back to Silver/Bronze sources.

5. Scope

- Six source datasets: customer, product, orders, store sales, inventory and payment.
- Bronze, Silver and Gold data layers.
- Data quality, deduplication, standardization and exception handling.
- Customer 360 and RFM segmentation.
- Sales, store, channel, inventory and payment analytics.
- Executive and operational dashboards.
- ML/AI readiness including churn, forecasting and GenAI analysis.
- Governance, security, lineage and auditability.

6. Functional Requirements

FR-01 — Ingestion: Ingest all six source datasets into Bronze with source metadata and incremental processing support.

FR-02 — Bronze Preservation: Retain source-aligned data for audit, replay and historical analysis.

FR-03 — Data Quality: Validate mandatory fields, data types, duplicates, relationships, quantities, amounts and dates.

FR-04 — Silver Transformation: Cleanse, standardize, deduplicate and integrate source data.

FR-05 — Gold Analytics: Create business-ready Gold datasets for agreed analytical domains.

FR-06 — Customer 360: Combine customer profile and transaction behavior into a unified view.

FR-07 — RFM Segmentation: Calculate recency, frequency and monetary metrics and assign customer segments.

FR-08 — Inventory Risk: Identify products at/below reorder levels and stock-out risk.

FR-09 — Channel Analytics: Compare online and store performance using consistent KPIs.

FR-10 — Payment Analytics: Reconcile payments with orders and identify success, refund and exception cases.

FR-11 — Dashboards: Provide filters and drill-downs by date, store, product, category and customer segment.

FR-12 — ML/AI: Provide curated datasets/features for churn, forecasting, recommendations and AI analysis.

FR-13 — Governance: Provide role-based access, lineage, ownership and audit controls.

7. Business KPIs and Insights

Area	KPIs / Insights
Sales	Revenue, orders, units sold, average order value, daily/monthly trend
Products	Top products, category contribution, product mix
Customers	Customer revenue, order frequency, AOV, RFM segment
Stores	Revenue, transactions, units, average transaction value, store ranking
Channels	Online vs store revenue and transaction contribution
Inventory	Available stock, reorder risk, safety-stock breach, stock-out risk
Payments	Payment value by method, success/refund rate, unmatched transactions
Customer Risk	At-risk customers based on recency, frequency and monetary behavior
Operations	Cancelled/refunded order value and exception counts

8. Data Quality Requirements

- Required business keys must be present for valid records.
- Duplicate orders, sales and payments must be detected.
- Quantity and amount validation must follow approved business rules.
- Foreign-key relationships must be monitored.
- Invalid records must be quarantined or flagged.
- Data-quality metrics must be measurable and monitored.

9. Non-Functional Requirements

Category	Requirement
Scalability	Support growth in transactions, customers, stores, products and sources.
Performance	Common dashboard queries should meet agreed interactive response targets.
Security	Apply least-privilege access and protect sensitive customer/payment data.
Auditability	Maintain pipeline logs, ingestion history and end-to-end lineage.
Recoverability	Support safe reruns without creating duplicate business transactions.
Maintainability	Use modular, reusable, parameterized pipelines and documented rules.

10. Target Databricks Solution

The target platform uses Databricks as the central Lakehouse platform. Medallion layers separate ingestion concerns from data-quality/integration concerns and business consumption concerns.

Architecture: Sources → Ingestion → Bronze → Quality & Integration → Silver → Business Rules & Aggregation → Gold → Databricks SQL / BI → ML / AI / Applications

Unity Catalog should provide governance across the governed data assets, including access control, ownership, discovery and lineage. Workflows/pipelines should orchestrate dependencies and provide operational visibility.

11. Analytics and AI Use Cases

- Sales analytics — revenue trends, product performance, store ranking and channel mix.
- Customer 360 — unified profile and transaction behavior.
- RFM segmentation — VIP, loyal, regular, dormant and at-risk customers.
- Inventory intelligence — low-stock and replenishment-risk identification.
- Churn prediction — identify customers likely to reduce or stop purchasing.
- Sales forecasting — forecast product/store demand.
- Product recommendations — identify complementary products.
- GenAI business analysis — answer questions such as why revenue changed for a store, product or period using governed Gold data.

12. Reporting Requirements

The executive dashboard should show revenue, orders, customers, AOV, store/channel performance, top products, inventory risk and major exceptions. Operational dashboards should support drill-down by date, store, product, category, customer segment and transaction status.

13. Security and Governance

- Classify and protect customer and payment-related information.
- Apply role-based access and least privilege.
- Protect sensitive columns according to organizational policy.
- Maintain lineage from source through Bronze, Silver and Gold.
- Audit important production data operations.

14. Acceptance Criteria

- All six source datasets are successfully ingested into Bronze.
- Bronze preserves source history and supports replay.
- Data-quality failures and duplicates are identified.
- Silver datasets are cleansed, standardized and integrated.
- Gold datasets support agreed KPIs and business insights.
- Customer 360 and RFM segmentation can be produced.
- Inventory reorder risks can be identified.
- Orders and payments can be reconciled.
- Dashboards expose required KPIs and filters.

- Governance and lineage are available across the Medallion layers.

15. Phased Implementation

Phase	Deliverables
1 — Foundation	Databricks workspace, governance, connectivity and Bronze ingestion.
2 — Engineering	Data quality, Silver transformations, deduplication and standardization.
3 — Analytics	Gold datasets, KPI definitions and dashboards.
4 — Customer & Inventory	Customer 360, RFM and inventory-risk analytics.
5 — ML	Churn prediction, sales forecasting and monitoring.
6 — GenAI	Natural-language business analysis using governed data.

16. Expected Business Benefits

The solution is expected to reduce manual reporting effort, improve trust in retail metrics, accelerate decision-making, identify inventory risks earlier, improve customer targeting, and establish a governed foundation for advanced analytics and AI.

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